

Health Workforce Modelling Approaches

Reflections from Sandon, SA

Joseph K. Mung'atu

July, 2025

Technical Working Report

Health Workforce Planning

Contents

1	Introduction	4
2	Modelling Approaches Overview	4
3	Input Variables for Health Workforce Models	4
3.1	Workforce Supply Variables	4
3.2	Population & Demand Variables	4
3.3	Health System & Policy Variables	5
3.4	Contextual & Economic Variables	5
4	Modelling Approaches & Their Mathematical Foundations	5
4.1	Stock-and-Flow Models	5
4.1.1	Basic Equation	5
4.1.2	Extended Formulation	6
4.1.3	Markov Chain Formulation	6
4.2	Needs-Based Models	6
4.2.1	Basic Needs Equation	6
4.2.2	Optimization Formulation	7
4.3	Utilization-Based Models	7
4.3.1	ARIMA Time Series Model	7
4.3.2	Simple AR(1) Model	8
4.3.3	Holt-Winters Model	8
4.4	Skill-Mix Models	8
4.4.1	Linear Programming Formulation	8
4.4.2	Integer Programming for Discrete Workforce	8
4.5	System Dynamics Models	9

4.5.1	Differential Equation Formulation	9
4.5.2	Feedback Loop Example	9
4.6	Agent-Based Models (ABM)	9
4.6.1	Agent State Transitions	9
4.6.2	Monte Carlo Simulation in ABM	9
4.7	Microsimulation Models	10
4.7.1	Discrete-Event Simulation	10
4.7.2	Markov Process for Individual Transitions	10
4.8	Cohort-Based Simulation	10
4.8.1	Matrix Projection Model	10
4.8.2	Survival Analysis Formulation	10
4.9	Optimization Models	10
4.9.1	Linear Programming Formulation	10
4.9.2	Goal Programming	11
4.9.3	Multi-Objective Optimization	11
4.10	Stochastic Models	11
4.10.1	Monte Carlo Simulation	11
4.10.2	Stochastic Differential Equation	11
4.11	Econometric Models	12
4.11.1	Multiple Regression	12
4.11.2	Poisson Regression for Count Data	12
4.11.3	Panel Data Model	12
4.12	Time-Series Models	12
4.12.1	Exponential Smoothing	12
4.12.2	Fourier Analysis for Seasonal Patterns	12
4.13	Workload Indicators of Staffing Need (WISN)	12

4.13.1 Basic WISN Calculation	13
4.13.2 Detailed Formulation	13
4.13.3 Ratio Calculations	13
4.14 Manpower Planning Models	13
4.14.1 Matrix-Based Projections	13
4.14.2 Markov Chain for Career Progression	13
5 Choosing the Right Model	13
5.1 Model Selection Criteria	13
5.2 Hybrid Approaches	14
6 Implementation Considerations	15
6.1 Data Requirements Matrix	15
6.2 Validation and Verification	15
7 Conclusion	15

1 Introduction

To model optimal health workforce numbers, experts use a range of sophisticated approaches that integrate both supply-side and demand-side variables. These models aim to forecast future needs, identify gaps, and support strategic planning. This document presents common health workforce modelling approaches along with the mathematical techniques typically used in each. These models help forecast supply, demand, and gaps in health workforce planning.

Health workforce modelling has evolved significantly over the past decades, moving from simple ratio-based calculations to complex, integrated systems that account for multiple interacting factors. The choice of modelling approach depends on several factors including data availability, planning horizon, policy questions, and computational resources available. In practice, many studies combine multiple modelling techniques to capture the full complexity of health systems.

2 Modelling Approaches Overview

Note: Many studies combine these models to capture the full complexity of health systems. The integration of multiple approaches allows for more robust and comprehensive workforce planning.

3 Input Variables for Health Workforce Models

3.1 Workforce Supply Variables

- **Current Workforce:** Numbers by cadre, specialty, and geographic location
- **Training Pipeline:** Training capacity, graduation rates, and education system capacity
- **Attrition & Exit:** Retirement rates, career change, mortality, and other forms of exit
- **Mobility:** Migration patterns (inward/outward flows), domestic mobility
- **Productivity:** Workforce productivity, hours worked, scope of practice, and skill mix
- **Regulation:** Professional regulation and licensing

3.2 Population & Demand Variables

- **Demographic Data:** Population size, growth rate, age structure, gender distribution, urban/rural split, fertility, and mortality rates

- **Health Status:** Disease burden, epidemiological trends, health indicators
- **Service Utilization:** Historical and projected service utilization rates
- **Coverage & Targets:** Universal Health Coverage (UHC) targets, service delivery targets
- **Epidemic Shocks:** Epidemic frequency, preparedness, and projected impact

3.3 Health System & Policy Variables

- **Health Financing:** Budget constraints, health expenditure, wage levels, economic factors
- **Policy Scenarios:** Strategic reforms, policy goals, regulation changes
- **System Reforms:** Health system delivery models, organizational structures
- **Technology & Innovation:** Technology adoption, task shifting, efficiency improvements
- **Education Investments:** Funding for training infrastructure and programs

3.4 Contextual & Economic Variables

- **Economic Factors:** Labor market conditions, wage levels, employment opportunities
- **Geographic Factors:** Distribution of facilities, rural/urban disparities, infrastructure
- **Temporal Factors:** Time delays in training, career progression timelines, policy implementation lags

4 Modelling Approaches & Their Mathematical Foundations

4.1 Stock-and-Flow Models

Stock-and-flow models track the movement of health workers through various states over time. They are fundamental for workforce supply projections.

4.1.1 Basic Equation

$$W_{t+1} = W_t + I_t - O_t \tag{1}$$

Where:

- W_t : Workforce at time t
- I_t : Total inflow (graduates, recruits, re-entrants)
- O_t : Total outflow (retirement, attrition, migration)

4.1.2 Extended Formulation

For multiple cadres and career stages:

$$\mathbf{W}(t+1) = \mathbf{A} \cdot \mathbf{W}(t) + \mathbf{G}(t) - \mathbf{L}(t) + \mathbf{M}(t) \quad (2)$$

Where:

- $\mathbf{W}(t)$: Vector of workforce stocks by cadre
- $\mathbf{A}$: Transition matrix (career progression)
- $\mathbf{G}(t)$: Graduation vector
- $\mathbf{L}(t)$: Loss vector (attrition, retirement)
- $\mathbf{M}(t)$: Net migration vector

4.1.3 Markov Chain Formulation

For probabilistic transitions:

$$P_{ij} = \mathbb{P}(\text{transition from state } i \text{ to state } j) \quad (3)$$

$$\mathbf{W}(t+1) = \mathbf{P}^\top \cdot \mathbf{W}(t) \quad (4)$$

4.2 Needs-Based Models

Needs-based models estimate workforce requirements based on population health needs and service targets.

4.2.1 Basic Needs Equation

$$D_k = \frac{P \times p \times \lambda \times C}{\mu} \quad (5)$$

Where:

- D_k : Required workers of cadre k
- P : Population size
- p : Disease prevalence or service need rate
- λ : Service frequency per case
- C : Coverage target (0-1)
- μ : Productivity (cases per worker)

4.2.2 Optimization Formulation

$$\min \sum_{i=1}^n (D_i - S_i)^2 \quad (6)$$

Where:

- D_i : Demand for cadre i
- S_i : Supply of cadre i

Subject to constraints:

$$\sum_{i=1}^n c_i S_i \leq B \quad (\text{Budget}) \quad (7)$$

$$S_i \geq S_i^{\min} \quad (\text{Minimum staffing}) \quad (8)$$

$$\frac{S_j}{S_i} \geq r_{ij} \quad (\text{Skill mix ratios}) \quad (9)$$

4.3 Utilization-Based Models

Utilization-based models project demand using historical service usage patterns.

4.3.1 ARIMA Time Series Model

$$\phi(B)(1 - B)^d Y_t = \theta(B)\epsilon_t \quad (10)$$

Where:

- $\phi(B)$: Autoregressive polynomial
- $\theta(B)$: Moving average polynomial
- B : Backshift operator

- d : Degree of differencing
- ϵ_t : White noise error term

4.3.2 Simple AR(1) Model

$$Y_t = \alpha + \beta Y_{t-1} + \epsilon_t \quad (11)$$

Useful for projecting workforce numbers based on historical trends.

4.3.3 Holt-Winters Model

For data with trend and seasonality:

$$\text{Level: } L_t = \alpha Y_t + (1 - \alpha)(L_{t-1} + T_{t-1}) \quad (12)$$

$$\text{Trend: } T_t = \beta(L_t - L_{t-1}) + (1 - \beta)T_{t-1} \quad (13)$$

$$\text{Seasonal: } S_t = \gamma(Y_t - L_{t-1} - T_{t-1}) + (1 - \gamma)S_{t-m} \quad (14)$$

4.4 Skill-Mix Models

Skill-mix models determine optimal combinations of different health worker types.

4.4.1 Linear Programming Formulation

$$\min \sum_{k=1}^K c_k x_k \quad (15)$$

Subject to:

$$\sum_{k=1}^K a_{ik} x_k \geq b_i \quad (\text{Service requirements}) \quad (16)$$

$$x_k \geq 0 \quad (\text{Non-negativity}) \quad (17)$$

$$\frac{x_j}{x_i} \geq r_{ij} \quad (\text{Minimum ratios}) \quad (18)$$

4.4.2 Integer Programming for Discrete Workforce

$$x_k \in \mathbb{Z}^+ \quad (\text{Integer workforce numbers}) \quad (19)$$

4.5 System Dynamics Models

System dynamics models capture feedback loops and delays in complex systems.

4.5.1 Differential Equation Formulation

$$\frac{dW}{dt} = I(t) - O(t) \quad (20)$$

Where flows are functions of stocks and other variables.

4.5.2 Feedback Loop Example

$$\frac{dW}{dt} = \alpha \cdot (D - W) - \beta \cdot W \quad (21)$$

$$\frac{dD}{dt} = \gamma \cdot P(t) - \delta \cdot D \quad (22)$$

Where workforce adjusts to meet demand, which itself changes with population.

4.6 Agent-Based Models (ABM)

ABMs simulate individual agents and their interactions.

4.6.1 Agent State Transitions

$$S_i(t+1) = f(S_i(t), E_i(t), \sum_{j \neq i} g(S_i(t), S_j(t))) \quad (23)$$

Where:

- $S_i(t)$: State of agent i at time t
- $E_i(t)$: External environment
- $g(\cdot)$: Interaction function with other agents

4.6.2 Monte Carlo Simulation in ABM

$$X_{i,t+1} \sim F(\theta_i, S_i(t), E_i(t)) \quad (24)$$

Where F is a probability distribution governing agent behavior.

4.7 Microsimulation Models

Microsimulation models simulate individual-level transitions.

4.7.1 Discrete-Event Simulation

$$T_{next} = \min\{t_{event}^1, t_{event}^2, \dots, t_{event}^n\} \quad (25)$$

Where t_{event}^i are times of next scheduled events for each individual.

4.7.2 Markov Process for Individual Transitions

$$\mathbb{P}(S_{i,t+1} = j \mid S_{i,t} = k) = p_{kj} \quad (26)$$

4.8 Cohort-Based Simulation

Cohort models track groups of workers over time.

4.8.1 Matrix Projection Model

$$\mathbf{n}(t+1) = \mathbf{A} \cdot \mathbf{n}(t) \quad (27)$$

Where $\mathbf{n}(t)$ is the cohort size vector and $\mathbf{A}$ is the transition matrix.

4.8.2 Survival Analysis Formulation

$$S(t) = \exp\left(-\int_0^t h(s)ds\right) \quad (28)$$

Where $h(s)$ is the hazard rate (attrition rate).

4.9 Optimization Models

Optimization models identify optimal workforce allocations under constraints.

4.9.1 Linear Programming Formulation

$$\max Z = \sum_{i=1}^n c_i x_i \quad (29)$$

Subject to:

$$\sum_{j=1}^n a_{ij}x_j \leq b_i \quad \text{for } i = 1, \dots, m \quad (30)$$

$$x_j \geq 0 \quad \text{for } j = 1, \dots, n \quad (31)$$

4.9.2 Goal Programming

$$\min \sum_{k=1}^K w_k |d_k^+ - d_k^-| \quad (32)$$

Where d_k^+, d_k^- are positive and negative deviations from goals.

4.9.3 Multi-Objective Optimization

$$\min \{f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_p(\mathbf{x})\} \quad (33)$$

Subject to $\mathbf{x} \in \mathcal{X}$.

4.10 Stochastic Models

Stochastic models incorporate uncertainty through probabilistic methods.

4.10.1 Monte Carlo Simulation

$$\hat{\theta} = \frac{1}{N} \sum_{i=1}^N f(X_i) \quad (34)$$

Where $X_i \sim P(x)$ are random samples from probability distributions of key parameters.

4.10.2 Stochastic Differential Equation

$$dW(t) = \mu(W, t)dt + \sigma(W, t)d\mathcal{B}(t) \quad (35)$$

Where:

- $\mu(W, t)$: Drift term (deterministic trend)
- $\sigma(W, t)$: Diffusion term (volatility)
- $\mathcal{B}(t)$: Brownian motion

4.11 Econometric Models

Econometric models analyze relationships between workforce variables.

4.11.1 Multiple Regression

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + \epsilon_i \quad (36)$$

4.11.2 Poisson Regression for Count Data

$$\log(\mathbb{E}[Y_i]) = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_k X_{ki} \quad (37)$$

For modelling workforce counts or events.

4.11.3 Panel Data Model

$$Y_{it} = \alpha_i + \beta X_{it} + \epsilon_{it} \quad (38)$$

Where α_i captures individual (or region) specific effects.

4.12 Time-Series Models

Time-series models forecast future workforce trends based on historical data.

4.12.1 Exponential Smoothing

$$\hat{Y}_{t+1} = \alpha Y_t + (1 - \alpha) \hat{Y}_t \quad (39)$$

4.12.2 Fourier Analysis for Seasonal Patterns

$$Y_t = a_0 + \sum_{k=1}^K [a_k \cos(2\pi kt/T) + b_k \sin(2\pi kt/T)] + \epsilon_t \quad (40)$$

4.13 Workload Indicators of Staffing Need (WISN)

WISN is a WHO-recommended approach for workload-based staffing.

4.13.1 Basic WISN Calculation

$$\text{WISN} = \frac{\text{Total Workload}}{\text{Available Working Time}} \times \text{Allowance Standard} \quad (41)$$

4.13.2 Detailed Formulation

$$\text{Required Staff} = \frac{\sum(\text{Activity Standards} \times \text{Workload})}{\text{Available Time}} \times (1 + \text{Category Allowance}) \quad (42)$$

4.13.3 Ratio Calculations

$$\text{Staffing Ratio} = \frac{\text{Actual Staff}}{\text{WISN}} \quad (43)$$

4.14 Manpower Planning Models

Manpower planning models align workforce supply with long-term service needs.

4.14.1 Matrix-Based Projections

$$\mathbf{M}(t) = \mathbf{P} \cdot \mathbf{M}(t-1) + \mathbf{R}(t) - \mathbf{L}(t) \quad (44)$$

Where $\mathbf{M}(t)$ is manpower vector, $\mathbf{P}$ is promotion matrix.

4.14.2 Markov Chain for Career Progression

$$\pi_{t+1} = \pi_t \mathbf{T} \quad (45)$$

Where π_t is state probability vector, $\mathbf{T}$ is transition matrix.

5 Choosing the Right Model

5.1 Model Selection Criteria

1. Planning Horizon:

- Short-term: Time-series, econometric models
- Medium-term: Stock-and-flow, optimization

- Long-term: System dynamics, needs-based, manpower planning

2. Data Availability:

- Rich data: Complex models (ABM, microsimulation, econometric)
- Limited data: Simple models (ratios, WISN, basic projections)

3. Policy Questions:

- "What if?" scenarios: Simulation models (ABM, system dynamics)
- Optimal allocation: Optimization models
- Impact assessment: Econometric, system dynamics
- Workforce mix: Skill-mix models

4. Computational Resources:

- Limited: Simple calculations, spreadsheet models, WISN
- Advanced: Complex simulations, optimization, microsimulation

5.2 Hybrid Approaches

In practice, most real-world applications use hybrid approaches that combine multiple modelling techniques:

$$\text{Hybrid Model} = \alpha \cdot \text{Model}_1 + \beta \cdot \text{Model}_2 + \gamma \cdot \text{Model}_3 \quad (46)$$

Common hybrid combinations:

- **Stock-and-flow with optimization:** Project supply then optimize allocation
- **Needs-based with stochastic elements:** Estimate needs with uncertainty bounds
- **System dynamics with agent-based components:** Capture both macro feedback and micro behaviors
- **Econometric with time-series:** Model relationships with trend components
- **WISN with optimization:** Workload-based needs with constrained allocation

6 Implementation Considerations

6.1 Data Requirements Matrix

6.2 Validation and Verification

1. Model Validation:

$$\text{Validation Score} = 1 - \frac{\sum |\text{Predicted} - \text{Actual}|}{\sum |\text{Actual}|} \quad (47)$$

2. Sensitivity Analysis:

$$S_i = \frac{\partial Y}{\partial X_i} \cdot \frac{X_i}{Y} \quad (48)$$

Identify which input parameters most affect model outputs.

3. Scenario Testing:

- Base case (current trends continue)
- Optimistic scenario (best-case conditions)
- Pessimistic scenario (worst-case conditions)
- Policy intervention scenarios (specific policy changes)

4. Cross-Validation:

$$\text{CV Error} = \frac{1}{K} \sum_{k=1}^K \text{Error}_k \quad (49)$$

For time-series: use rolling window validation.

7 Conclusion

Health workforce modelling is a critical component of effective health system planning. This document has presented 14 distinct modelling approaches, each with its own mathematical foundations, data requirements, and appropriate applications. The choice of modelling approach depends on multiple factors including the planning horizon, available data, policy questions, and computational resources.

Recommendations from Sandon's experience:

1. Start with simple models and gradually increase complexity as data and capacity allow
2. Use hybrid approaches to capture different aspects of the workforce system
3. Validate models against historical data and expert judgment

4. Conduct sensitivity analysis to understand key drivers and uncertainties
5. Engage stakeholders throughout the modelling process to ensure relevance and uptake
6. Consider both supply-side and demand-side factors in comprehensive planning
7. Account for uncertainty through stochastic methods or scenario analysis
8. Align model complexity with decision-making needs

Future directions in health workforce modelling include greater integration of artificial intelligence and machine learning techniques, improved handling of uncertainty through advanced stochastic methods, better integration of economic and behavioral factors, and more user-friendly tools for policy-makers. The development of standardized modelling frameworks that can be adapted to different country contexts remains an important area for further work.

Table 1: Health Workforce Modelling Approaches and Their Applications

Model Type	Purpose	Mathematical Techniques
Stock-and-Flow Models	Track workforce inflows (training, recruitment) and outflows (retirement, migration).	Arithmetic calculations, differential equations, Markov chains
Needs-Based Models	Estimate workforce needs based on population health status and service targets.	Epidemiological modelling, service utilization rates, optimization
Utilization-Based Models	Project demand using historical service usage data.	Time-series analysis (e.g., ARIMA), regression modelling
Skill-Mix Models	Determine optimal combinations of health worker types.	Linear programming, integer optimization, simulation
System Dynamics Models	Simulate feedback loops and time delays in complex systems.	Differential equations, causal loop diagrams, stock-and-flow diagrams
Agent-Based Models (ABM)	Model individual agents (e.g., health workers, patients) and their interactions.	Stochastic processes, state-transition logic, Monte Carlo simulation
Microsimulation Models	Simulate individual-level transitions (e.g., career paths, migration).	Probabilistic modelling, discrete-event simulation, Markov processes
Cohort-Based Simulation	Track groups of workers over time.	Matrix algebra, survival analysis, transition probabilities
Optimization Models	Identify best-case workforce allocation under constraints.	Linear/nonlinear programming, goal programming, multi-objective optimization
Stochastic Models	Account for uncertainty and randomness (e.g., epidemics, migration shocks).	Monte Carlo simulation, probabilistic forecasting, stochastic differential equations
Econometric Models	Analyze relationships between workforce variables and outcomes.	Regression analysis (OLS, logistic, Poisson), panel data models
Time-Series Models	Forecast future trends based on historical data.	ARIMA, exponential smoothing, Holt-Winters, Fourier analysis

Table 2: Variable Usage by Modelling Approach

Modelling Approach	Priority Variables
Stock-and-flow models	Heavily rely on workforce supply variables
Needs-based models	Prioritize population health and demographic variables
Optimization models	Require detailed economic and constraint variables
Agent-based models	Incorporate individual-level behavioral variables not always captured in aggregate lists
Econometric models	Focus on relationships between workforce outcomes and explanatory variables
System dynamics	Require variables that capture feedback loops and time delays

Table 3: Model Selection Guidelines

Planning Context	Recommended Modelling Approaches
Short-term planning (1-2 years)	Time-series models, utilization models, ratio-based methods
Long-term strategic planning (5-10 years)	Stock-and-flow models, needs-based models, system dynamics
Policy simulation and impact assessment	Agent-based models, stochastic models, optimization models
Data-rich environments	Econometric models, microsimulation, machine learning approaches
Data-scarce settings	Ratio-based methods, WISN, simplified manpower planning
Regional allocation problems	Optimization models, location-allocation models
Workforce mix optimization	Skill-mix models, linear/integer programming
Uncertainty and risk analysis	Stochastic models, Monte Carlo simulation, scenario analysis
Complex system behavior	System dynamics, agent-based models, network models
Training pipeline planning	Cohort models, Markov chain models, simulation
Individual-level analysis	Agent-based models, microsimulation
Workload-based staffing	WISN, activity-based costing, task-time analysis
Economic analysis	Econometric models, cost-benefit analysis, regression models

Table 4: Data Requirements by Model Type

Model Type	Critical Data Requirements	Data Quality Needs
Stock-and-flow	Current workforce, attrition rates, training pipeline	High precision, longitudinal
Needs-based	Population data, disease burden, service standards	Accurate, comprehensive
Utilization-based	Historical service data, utilization rates	Consistent, time-series
Skill-mix	Cadre productivity, service requirements, costs	Detailed, comparable
System dynamics	Feedback parameters, time delays, stocks/flows	Causal relationships, time-series
Agent-based	Individual behaviors, interaction rules, environments	Detailed, behavioral
Microsimulation	Individual-level data, transition probabilities	Granular, longitudinal
Cohort-based	Cohort characteristics, survival rates, transitions	Group-level, consistent
Optimization	Cost data, constraints, objective parameters	Reliable, consistent
Stochastic	Probability distributions, uncertainty parameters	Well-characterized distributions
Econometric	Historical data, explanatory variables, outcomes	Clean, sufficient observations
Time-series	Time-ordered data, trend/seasonality patterns	Consistent intervals, sufficient length
WISN	Activity standards, workload measurements, time data	Standardized, measurable
Manpower planning	Career structures, promotion patterns, exits	Structural, longitudinal